



HACKTHEBOX

Op SalwarKameez 5



OpSalwarKameez 5 - Bling-Bling

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Difficulty: **Easy**

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Scenario

It's the festive season of Diwali, and a newly launched website, Bling-Bling Crackers (a fictional site, inspired by platforms like Amazon and Flipkart), is offering huge discounts and free gifts for new users. To celebrate the festival of lights, Deepam Mart has launched a "Diwali Dhamaka" sale, offering ₹1 products, 50% off discounts, and free gifts like Diwali decorations and sweets for first-time users. The company Bling-Bling Crackers uses StoreD.

Artifacts Provided

- bf470e205234a1312c94e6ee3f6d648c artifacts/opsk-5.zip

Skills Learned

- Data Analysis
- Query Construction

Tags

- Neo4j
- Data Leak

Initial Analysis

To set up the Neo4j database, you can either install it locally on your computer or use Neo4j Aura, a cloud-based option.

Neo4j Aura offers a free instance, which can be used to complete this exercise without needing to install Neo4j yourself. Once your instance is set up, you can import the database via the "Import Database" option.

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Instance01

■ Restoring

...

Open

Instance ID **56bcf210** Neo4j version **5** Region **Belgium (europe-west1)** Nodes **2802 / 200000 (1%)** Relationships **1406 / 400000 (0%)**

Connection URI **neo4j+s://56bcf210.databases.neo4j.io**

Snapshots

Import Database

Logs

Drag & Drop your Neo4j **.dump** or **.backup** or **.tar** file here (4GB or less)
or [browse](#)

Want to upload a dump or backup larger than 4GB? [Click here](#) to view the Neo4j Admin 'upload' command in the Aura Docs.
The graph you upload should be smaller than 200,000 nodes and 400,000 relationships.

After importing, open the instance to start your analysis.



Database information

Nodes (2,802)

*

Account

CreditCard

Relationships (1,406)

*

HAS_CREDITCARD

Property keys

address

card_number

created_at

cvc

email

expiry_date

holder_name

phone_number

register_ip_address

username

1. Total number of Nodes in the database?

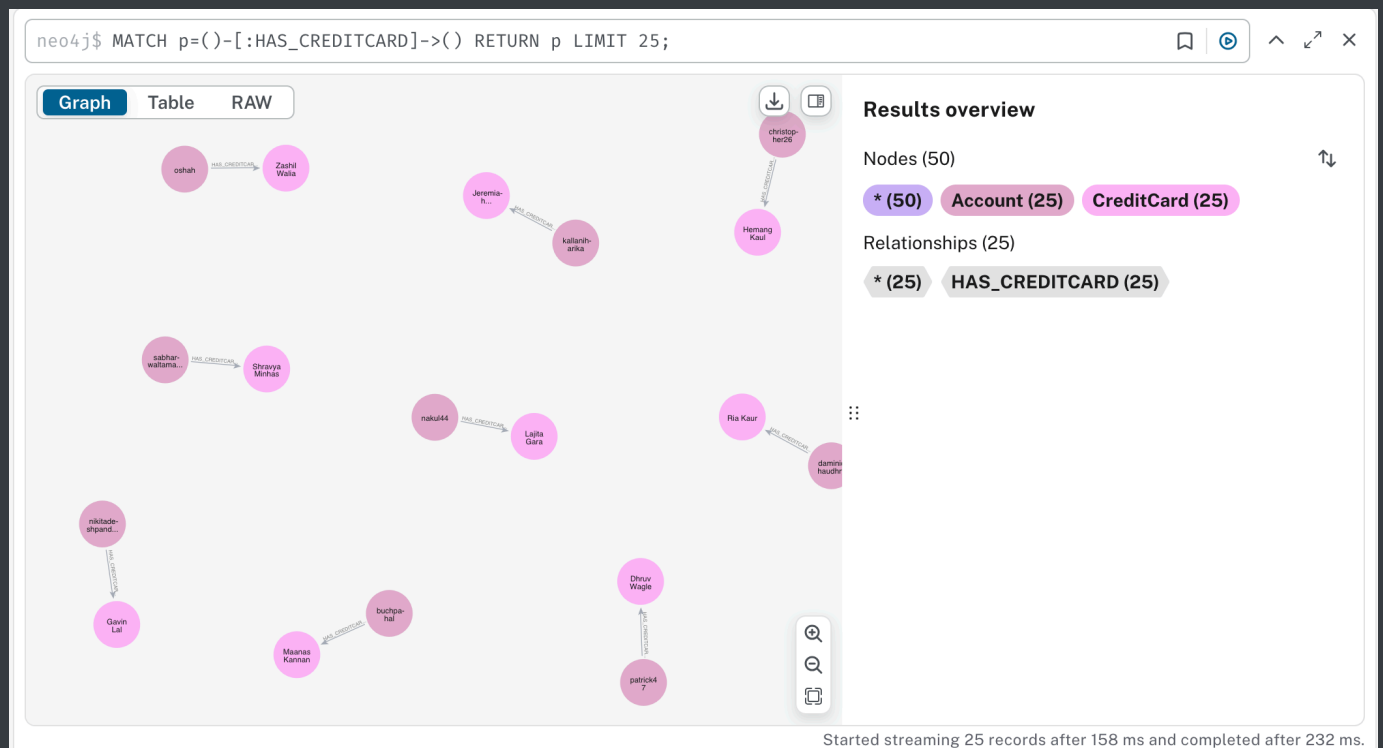
The database information panel displays the total number of nodes and relationships. However, you can also retrieve this information using the following query:

```
MATCH (n) RETURN count(n);
```

Answer: 2802

2. How many Account nodes does the database have?

The relationship data reveals that Account nodes are connected to Credit Card nodes, which will be useful for later queries.



To find the total number of Account nodes, you can check the database information or use this query:

```
MATCH (n:Account) RETURN count(n);
```

Answer: 1406

3. How many accounts are registered from the IP address 88.236.1.190?

Each Account node has several properties, including `register_ip_address`, which records the IP address used during account registration. Use the following query to find the count of accounts registered with a specific IP address:

```
match (n:Account) where n.register_ip_address = "88.236.1.190" return count(n)
```

```
neo4j$ match (n:Account) where n.register_ip_address = "88.236.1.190" return count(n)
```

count(n)

1 22

Started streaming 1 record after 54 ms and completed after 73 ms.

The 22 accounts registered with this IP might suggest a single owner or could indicate shared usage through NAT networking.

Answer: 22

4. How many Users have created multiple accounts with same IP address?

To identify other accounts created from the same IP address (not just the specific IP given), use this query to detect potential fraudulent activities:

```
MATCH (n:Account)
WITH n.register_ip_address AS prop1, COLLECT(n) AS nodes
WHERE SIZE(nodes) > 1
RETURN prop1, SIZE(nodes) AS duplicate_count
```

```
1 MATCH (n:Account)
2 WITH n.register_ip_address AS prop1, COLLECT(n) AS nodes
3 WHERE SIZE(nodes) > 1
4 RETURN prop1, SIZE(nodes) AS duplicate_count
```

prop1 duplicate_count

1 "88.236.1.190" 22

2 "149.116.170.185" 16

Started streaming 2 records after 83 ms and completed after 97 ms.

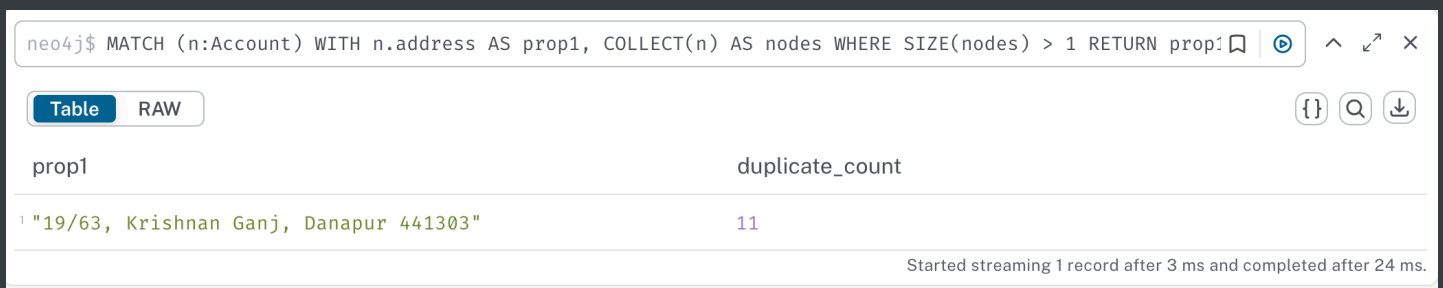
$$22 + 16 = 38$$

Answer: 38

5. What physical address has been used multiple times?

We can also investigate shared physical addresses, which are more challenging to change than IP addresses due to delivery requirements. Use the following query:

```
MATCH (n:Account)
WITH n.address AS prop1, COLLECT(n) AS nodes
WHERE SIZE(nodes) > 1
RETURN prop1, SIZE(nodes) AS duplicate_count
```



The image shows a screenshot of the Neo4j query interface. At the top, a query bar contains the Cypher query: `neo4j$ MATCH (n:Account) WITH n.address AS prop1, COLLECT(n) AS nodes WHERE SIZE(nodes) > 1 RETURN prop1`. Below the query bar, there are two tabs: "Table" (selected) and "RAW". To the right of the tabs are icons for JSON, search, and download. The results are displayed in a table with two columns: "prop1" and "duplicate_count". The first row shows the address `"19/63, Krishnan Ganj, Danapur 441303"` and the count `11`. At the bottom right, a status message reads: "Started streaming 1 record after 3 ms and completed after 24 ms."

prop1	duplicate_count
"19/63, Krishnan Ganj, Danapur 441303"	11

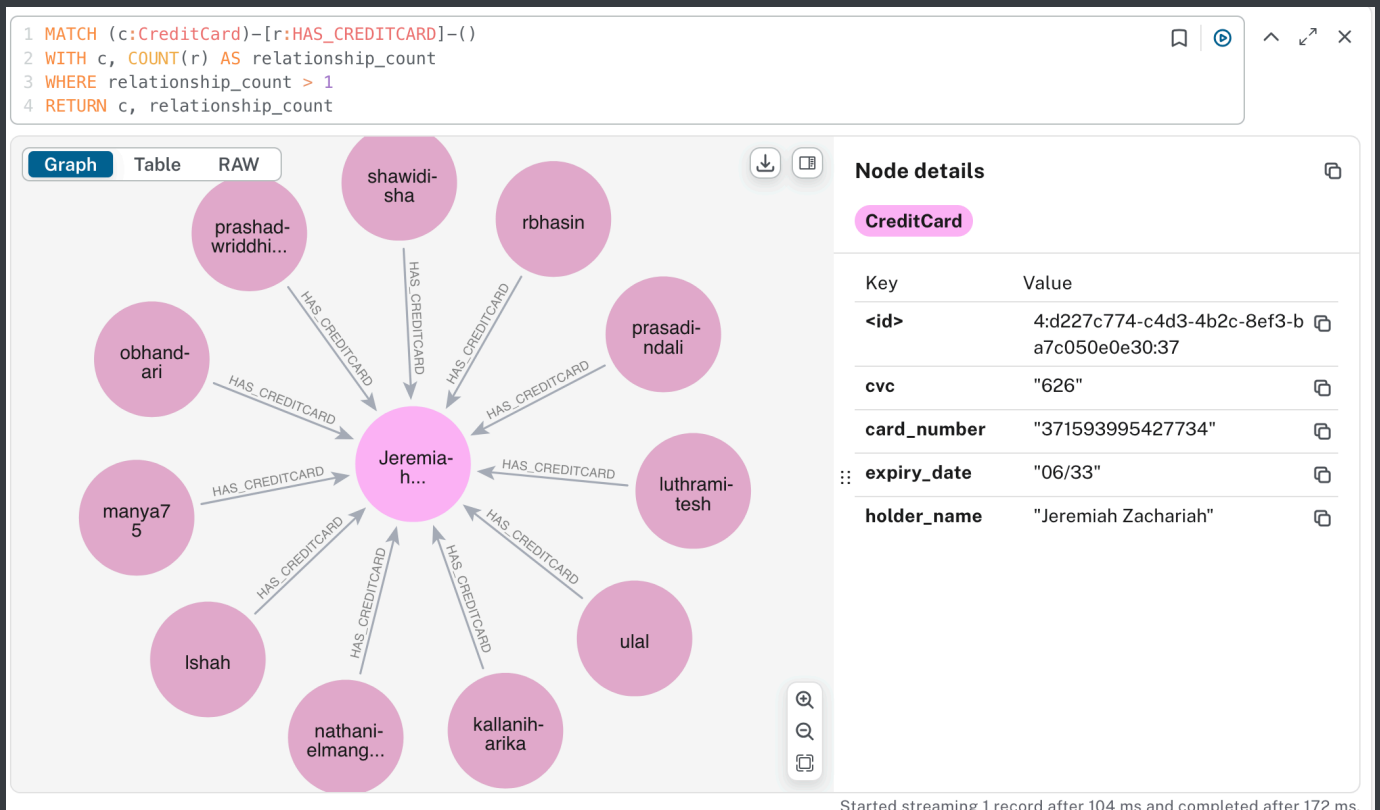
"19/63, Krishnan Ganj, Danapur 441303"

Answer: 19/63, Krishnan Ganj, Danapur 441303

6. Which Credit Card number is attached to multiple accounts?

Additionally, to detect potential credit card fraud (e.g., a single credit card number being associated with multiple accounts), use the query below:

```
MATCH (c:CreditCard)-[r:HAS_CREDITCARD]-()
WITH c, COUNT(r) AS relationship_count
WHERE relationship_count > 1
RETURN c, relationship_count
```



Answer: 371593995427734

7. When was the account with username obhandari created?

To determine when a specific account was created, use this query to see if the creation date aligns with any suspicious activity:

```
match (m:Account) where m.username = "obhandari" return m.created_at
```

Answer: 2024-11-08 07:42:20.228775

8. How many accounts using the credit card number from question 6 use the same registered IP address?

Finally, if you need to check the number of accounts using the same credit card and registered from the same IP address, the following query provides the count:

```
MATCH (c:CreditCard)-[r:HAS_CREDITCARD]-()
WITH c, COUNT(r) AS relationship_count
WHERE relationship_count > 1
RETURN relationship_count
```

Answer: 11